

ARDS Prediction Model

2021170850 최준서

2021170861 이석빈

2022170856 이호성

2023170818 장한이

2023170839 이영선

CONTENTS

Chapter 01. Preprocessing

Chapter 02. Prediction task

ARDS(급성 호흡곤란 증후군)

Chapter 03. Annotation & Labeling

Chapter 04. Model Design & Training

1. Machine Learning

2. Deep Learning

Chapter 05. Performance comparison and conclusion

+ References

Ch 01. Preprocessing



Core of Analyzing Released DF : Defining what to analyze and ensuring data quality

- Total Cohort : 11,514 unique ids secured.
- Inclusion Criteria : At least 2 days of hospitalization ($Los \geq 2$)
- Reason : To stably learn the pattern of early acute respiratory distress syndrome (ARDS) by securing a sufficient observation window of 48 hours.
- Strictly adhere to the Train 80%/Valid 10%/Test 10% ratio as defined in 'split_stay_id.json' to ensure the generalization performance of the model. (The criteria will be used on "prediction task choosing")



Analyzing Released DF

	charttime	label	value	stay_id	SOFA	los	Survival	In-hospital_death
0	00:00	Age	54.0	132539	1	5	-1	0
1	00:00	Gender	0.0	132539	1	5	-1	0
2	00:00	Height	NaN	132539	1	5	-1	0
3	00:00	ICUType	4.0	132539	1	5	-1	0
4	00:00	Weight	NaN	132539	1	5	-1	0
...	...	...	...	...	...	...	...	...
5085005	47:27	DiasABP	47.0	163037	7	21	-1	0
5085006	47:27	HR	74.0	163037	7	21	-1	0
5085007	47:27	MAP	63.0	163037	7	21	-1	0
5085008	47:27	SysABP	99.0	163037	7	21	-1	0
5085009	47:27	Urine	30.0	163037	7	21	-1	0

5085010 rows × 8 columns



label			
HR	662451	Creatinine	40708
MAP	426092	BUN	40527
SysABP	425439	Mg	39718
DiasABP	425314	Na	39691
Urine	397332	HCO3	39666
Weight	372097	Glucose	38159
NISysABP	283574	WBC	37791
NIDiasABP	283305	Lactate	23380
NIMAP	279114	SaO2	23345
Temp	251802	Age	11514
GCS	178582	Gender	11514
RespRate	159120	ICUType	11514
FiO2	91355	Height	11514
MechVent	87762	Bilirubin	9438
pH	69794	AST	9336
PaCO2	66685	ALT	9333
PaO2	66520	ALP	9087
HCT	53443	Albumin	6963
K	42454	TroponinT	6029
Platelets	41441	TroponinI	1155
		Cholesterol	952



Analyzing Released DF

label			
HR	662451	Creatinine	40708
MAP	426092	BUN	40527
SysABP	425439	Mg	39718
DiasABP	425314	Na	39691
Urine	397332	HC03	39666
Weight	372097	Glucose	38159
NISysABP	283574	WBC	37791
NIDiasABP	283305	Lactate	23380
NIMAP	279114	SaO2	23345
Temp	251802	Age	11514
GCS	178582	Gender	11514
RespRate	159120	ICUType	11514
FiO2	91355	Height	11514
MechVent	87762	Bilirubin	9438
pH	69794	AST	9336
PaCO2	66685	ALT	9333
PaO2	66520	ALP	9087
HCT	53443	Albumin	6963
K	42454	TroponinT	6029
Platelets	41441	TroponinI	1155
		Cholesterol	952

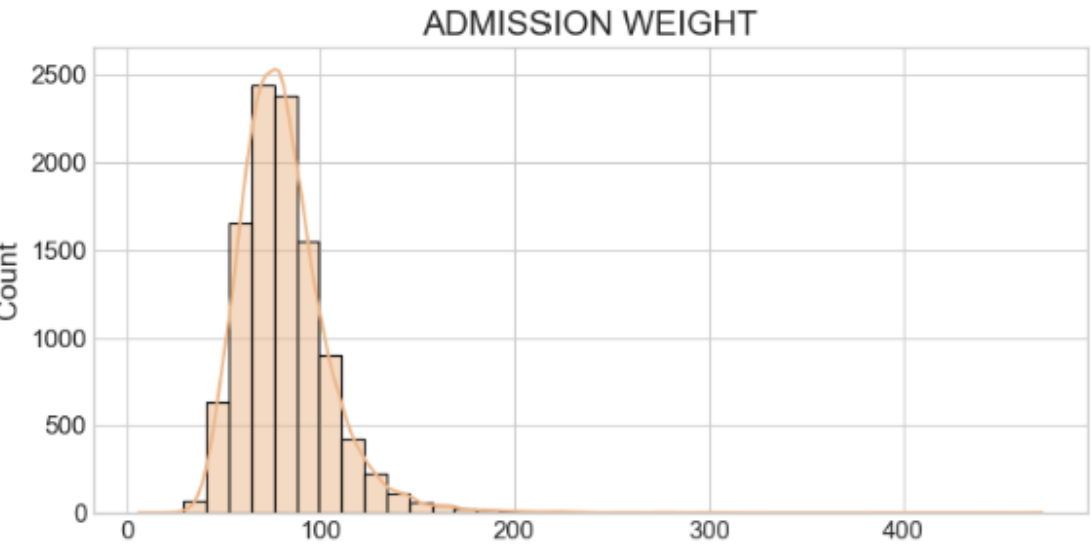
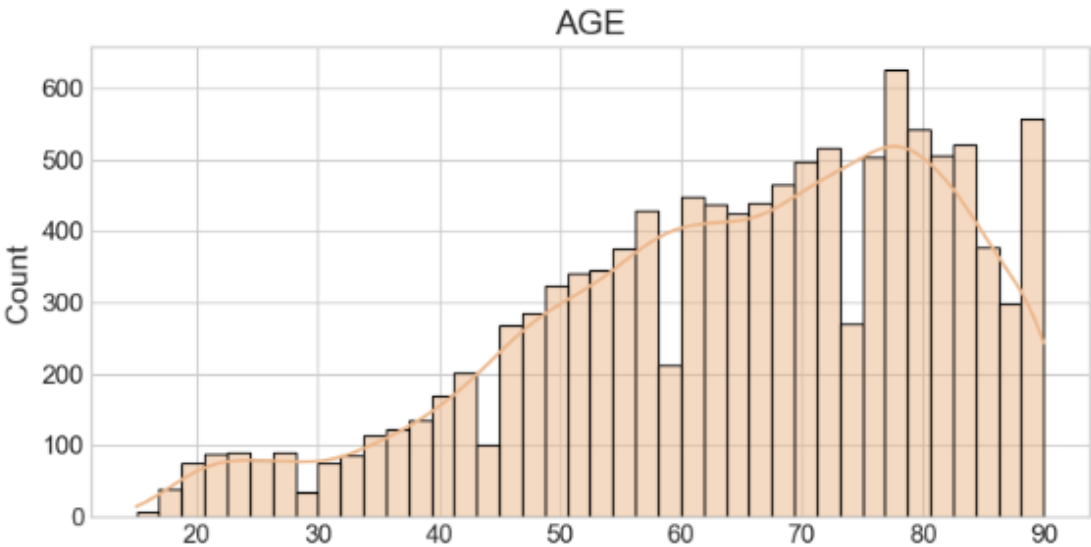
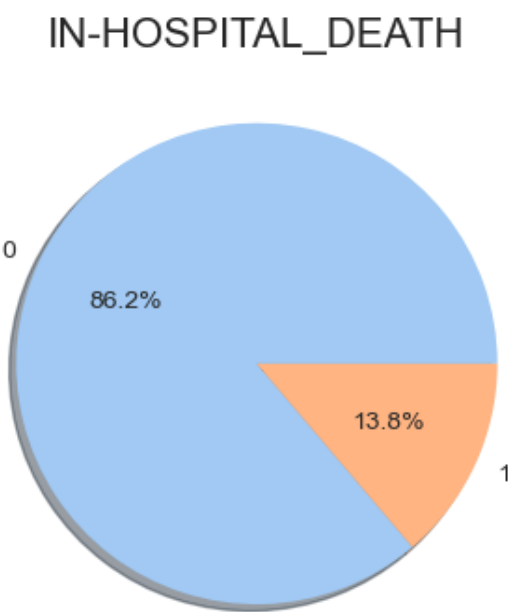
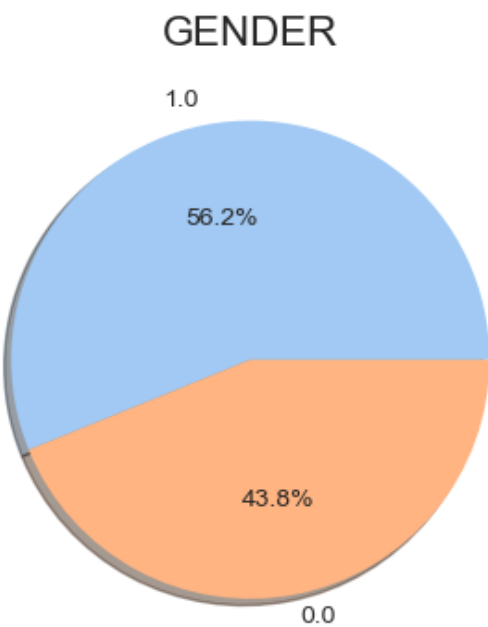
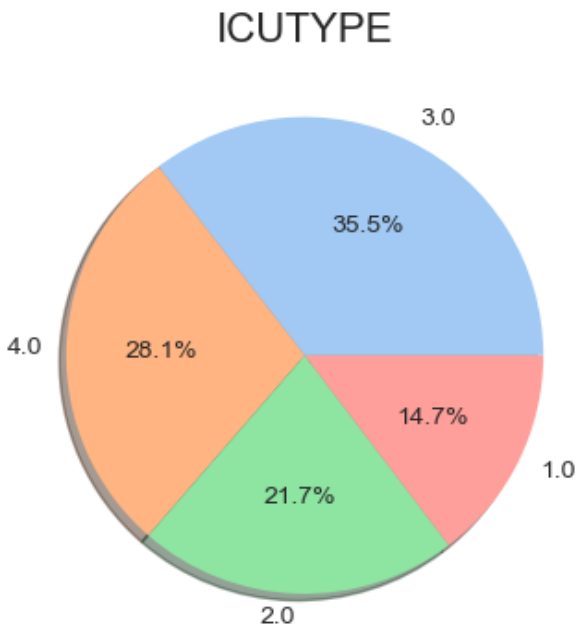
Static Variable
→ Age, Gender, ICUType, Height, Weight

Time - Variable
→ Albumin, RespRate...



Analyzing Released DF

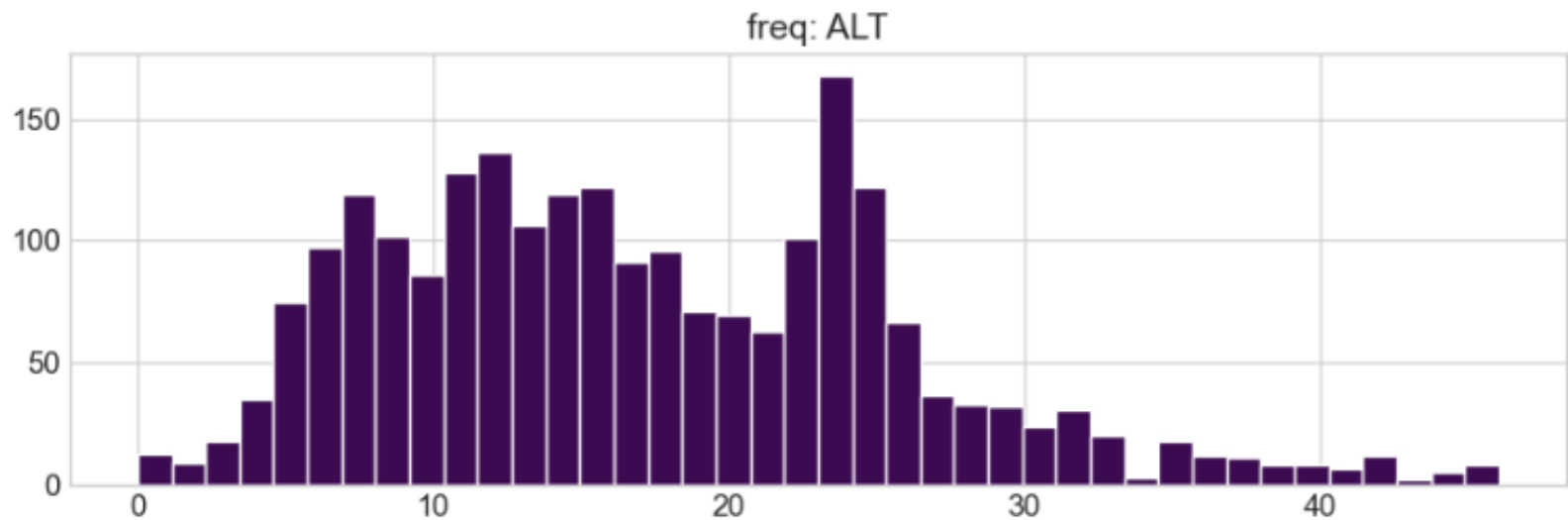
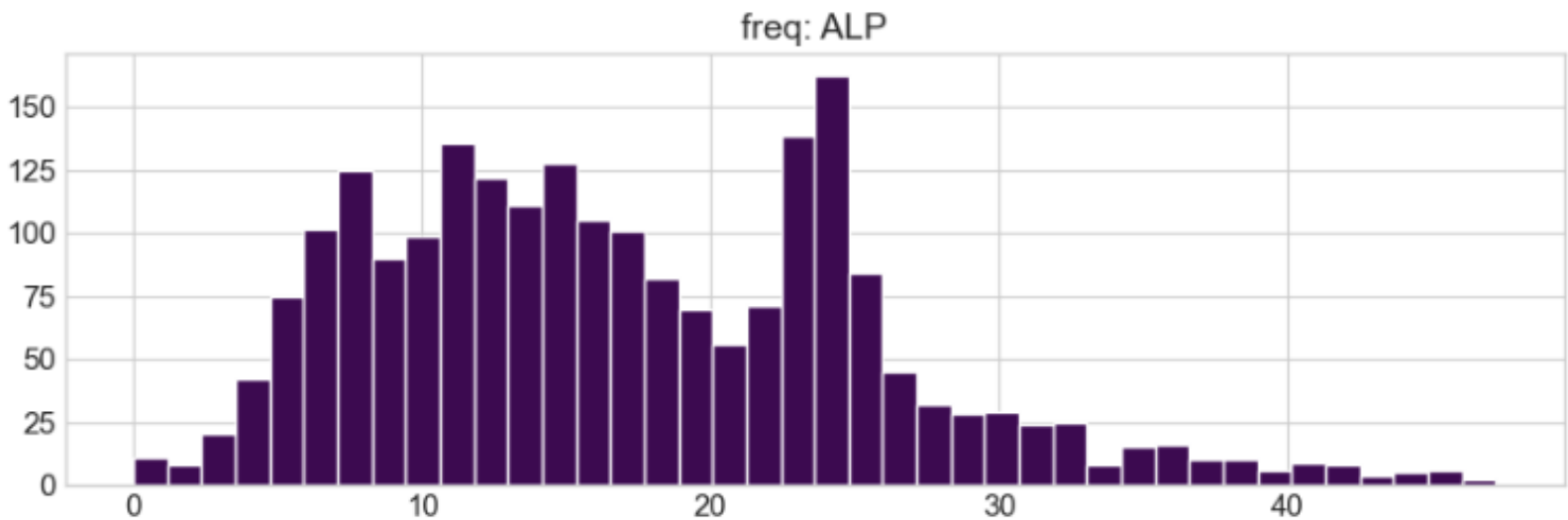
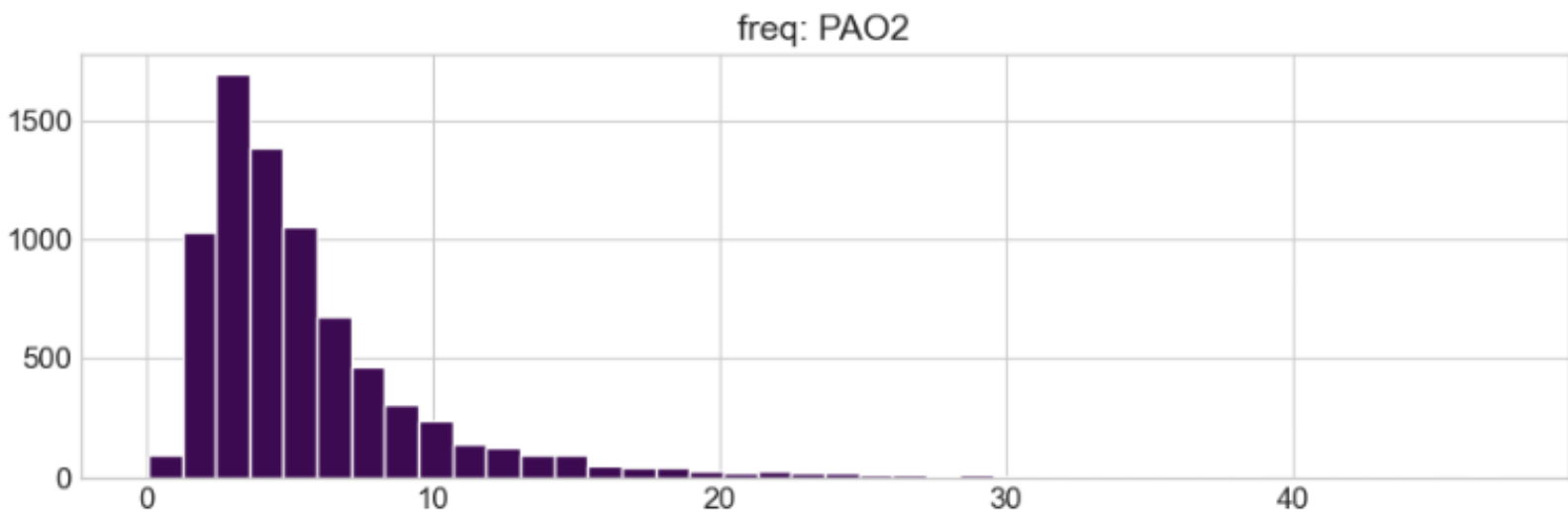
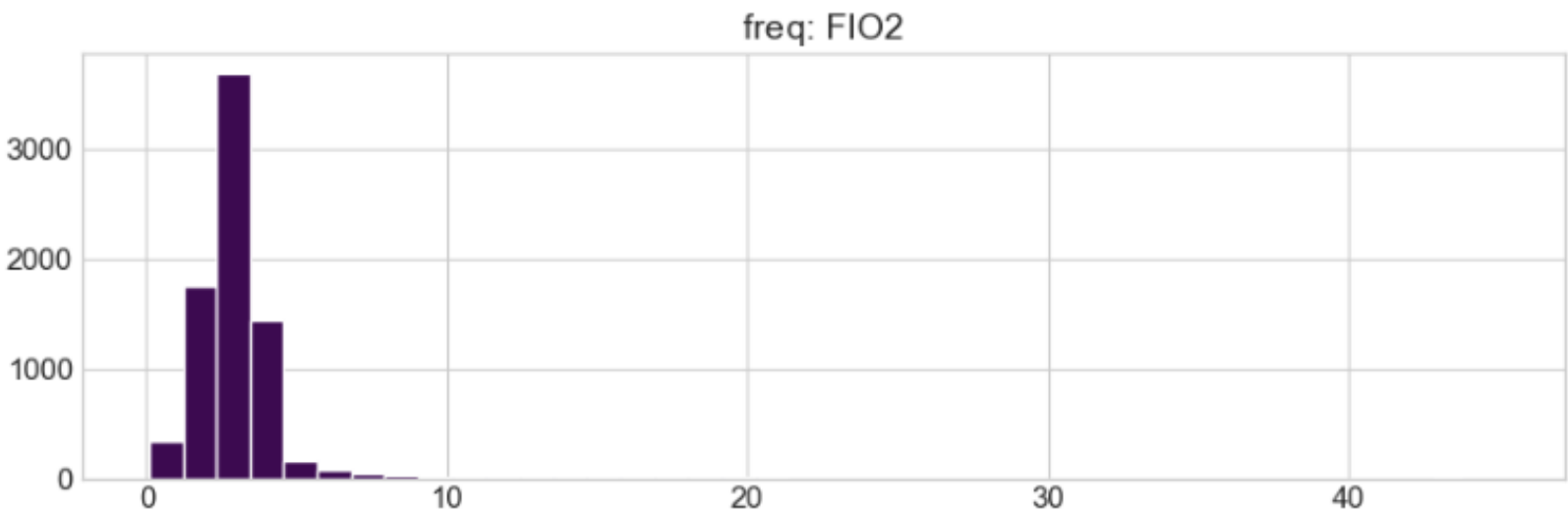
Static Variables with Pie chart / Histogram





Analyzing Released DF

Time – variable (dynamic) with Frequency



Outlier criteria

Outlier criteria is selected using domain knowledge.

MIMIC-Extract: A Data Extraction, Preprocessing, and Representation Pipeline for MIMIC-III

Shirly Wang
Layer 6 AI, University of Toronto
shirlywang@cs.toronto.edu

Matthew B.A. McDermott
Massachusetts Institute of Technology
mmd@mit.edu

Geeticka Chauhan
Massachusetts Institute of Technology
geeticka@mit.edu

Marzyeh Ghassemi
University of Toronto, Vector Insitute
marzyeh@cs.toronto.edu

Michael C. Hughes
Tufts University
mhughes@cs.tufts.edu

Tristan Naumann
Microsoft Research
tristan@microsoft.com

ABSTRACT

Machine learning for healthcare researchers face challenges to progress and reproducibility due to a lack of standardized processing frameworks for public datasets. We present MIMIC-Extract, an open source pipeline for transforming the raw electronic health record (EHR) data of critical care patients from the publicly-available MIMIC-III database into data structures that are directly usable in common time-series prediction pipelines. MIMIC-Extract addresses three challenges in making complex EHR data accessible to the broader machine learning community. First, MIMIC-Extract transforms raw vital sign and laboratory measurements into usable hourly time series, performing essential steps such as unit con-

April 2–4, 2020, Toronto, ON, Canada. ACM, New York, NY, USA, 14 pages.
<https://doi.org/10.1145/3368555.3384469>

INTRODUCTION

Applying modern machine learning to observational health data holds the potential to improve healthcare in many ways, such as delivering better patient treatments, improving hospital operations, and answering fundamental scientific questions [8]. To realize this potential, there have been efforts to make healthcare data available to credentialed researchers with human subjects training. A widely-used public data source is the Medical Information Mart for Intensive Care (MIMIC-III) dataset [14], which makes available the



Matthew B.A. McDermott
Massachusetts Institute of Technology mmd@mit.edu
→ The author of the paper - MIMIC-Extract
(The referenced criteria are in the author’s web-page)

Outlier criteria

Physiologically impossible values

```
outlier_limits = {  
    'Albumin': (0,60), 'ALP':(0,4000), 'ALT':(0,10000), 'AST':(0,10000), 'Bilirubin':(0,60),  
    'BUN':(0,250), 'Cholesterol':(0,1000), 'Creatinine':(0,66), 'DiasABP': (0,375), 'Glucose': (0,2200),  
    'HCO3':(0,66), 'HCT':(0,100), 'HR': (0,390), 'K':(0,15), 'Lactate':(0,33), 'Mg':(0,22),  
    'MAP': (0,375), 'Na':(0,250), 'NIDiasABP':(0,300), 'NIMAP':(0,300), 'NISysABP':(0,400),  
    'PaCO2':(0,220), 'PaO2':(0,770), 'pH': (6.3,10), 'Platelets':(0,2200), 'RespRate': (0,330),  
    'SaO2':(0,150), 'SysABP': (0,375), 'Temp': (14,46), 'TroponinI':(0,575), 'Troponin-T':(0,24),  
    'Urine':(0,2445), 'WBC':(0,1100)  
}
```

- 'FiO2', 'GCS', 'MechVent', 'Age', 'Gender', 'ICUType'
→ excluded from detection due to the nature of the variable. (Inadequate form such as Binary)

Outlier criteria

range is replaced with the nearest valid value. In generating the default cohort, we replace 35,251 (0.05%) measurements classified as non-valid outliers with nearest valid values and remove 5,402 (0.008%) measurements classified as extreme outliers. Appendix A lists the proportion of outliers detected at an aggregated feature level.

→ In the above paper 225p. same domain, 0.008% of MIMIC-III datasets are excluded as extreme outliers.

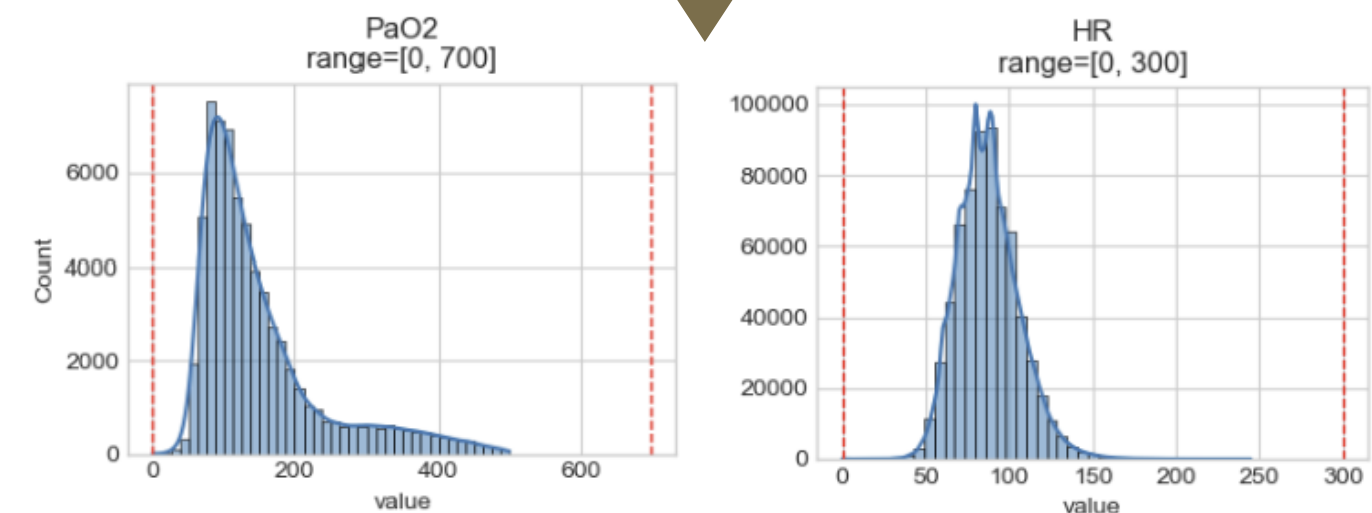
도메인 기준 이상치 제거
원래 행 개수: 5085010
제거된 행 개수: 661

→ Based on a domain basis : 0.013% removed

Std dev 기준 이상치 제거
 $\pm 1\sigma$, 제거된 행 개수: 1087678
 $\pm 2\sigma$, 제거된 행 개수: 174178
 $\pm 3\sigma$, 제거된 행 개수: 43700

→ Based on standard deviations: 0.8% removed

⇒ We adopt criteria based on domain knowledge!



Missing Value Imputation

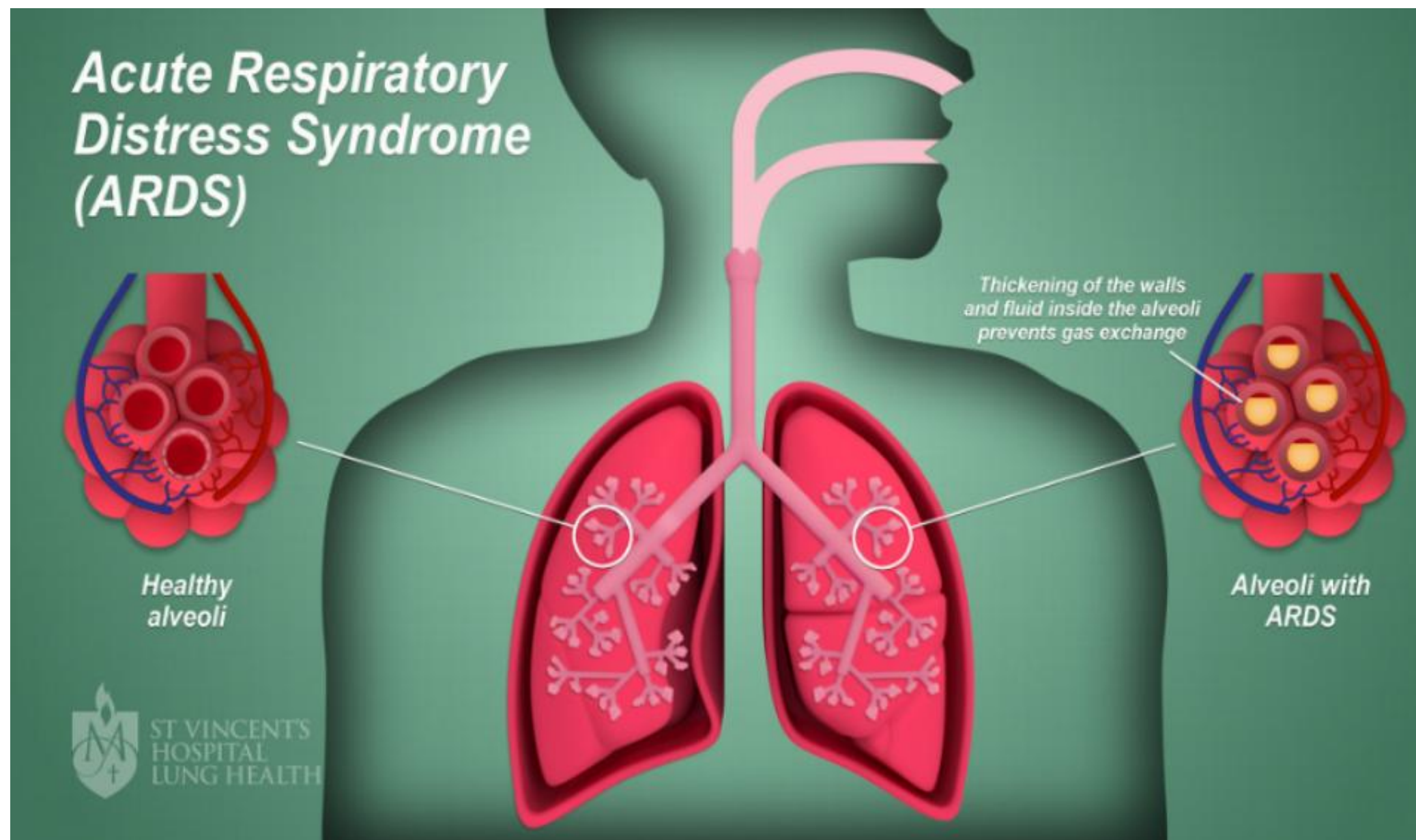
```
static_cols = [ 'stay_id', 'timestamp', 'SOFA', 'Survival',  
                'In-hospital_death', 'los', 'ICUType' ]
```

```
exclude_cols = [ 'MechVent', 'Height ', 'Gender' ]
```

- Fill NaNs with 0 for binary variables and non-imputable height.
- Other columns [`set(df.columns) - set(static_cols) - set(exclude_cols)`] : filled to zero by 'forward fill zero' method.

Ch 02. Prediction task

ARDS?



ARDS (Acute Respiratory Distress Syndrome) is a severe, life-threatening lung condition where fluid leaks into the tiny air sacs (alveoli), preventing enough oxygen from reaching the blood, causing severe shortness of breath and dangerously low blood oxygen levels, often in critically ill patients already in the hospital from infections or injuries. It's a rapid deterioration requiring immediate medical help, characterized by stiff lungs, fast breathing, and confusion.

WHY NPS-ARDS?

Machine learning-based model for predicting the occurrence and mortality of nonpulmonary sepsis-associated ARDS

Jinfeng Lin^{1,2}, Chunfeng Gu³, Zhaorui Sun¹, Suyan Zhang^{2✉} & Shinan Nie^{1✉}

Objective: The objective was to establish a machine learning-based model for predicting the occurrence and mortality of nonpulmonary sepsis-associated ARDS. **Methods:** 80% of sepsis patients selected randomly from the MIMIC-IV database, without prior pulmonary conditions and with nonpulmonary infection sites, were used to construct prediction models through machine learning techniques (including K-nearest neighbour, extreme gradient boosting, support vector machine, deep neural network, and decision tree methods). The remaining 20% of patients were utilized to validate the model's accuracy. Additionally, local data were employed for further model validation. **Results:** A total of 11,409 patients were included, with the most common type of infection being bloodstream infection. A total of 7,632 (66.9%) patients developed nonpulmonary sepsis-associated ARDS (NPS-ARDS). Patients with NPS-ARDS had significantly longer ICU stays (6.2 ± 5.2 days vs. 4.4 ± 3.7 days, $p < 0.01$) and higher 28-day mortality rates (19.5% vs. 14.9%, $p < 0.01$). Both internal and external validation demonstrated that the model constructed with the extreme gradient boosting method had high accuracy. In the internal validation, the model predicted NPS-ARDS and mortality in such patients with accuracies of 77.5% and 71.8%, respectively. In the external validation, the model predicted NPS-ARDS and mortality in these patients with accuracies of 78.0% and 81.4%, respectively. **Conclusion:** The model established via the extreme gradient boosting method can predict the occurrence and mortality of nonpulmonary sepsis-associated ARDS to a certain extent.

Keywords Machine learning, Predictive model, Sepsis, Acute respiratory distress syndrome, Mortality, Internal validation, External validation

- **Clinical Impact**
: Higher mortality (19.5% vs 14.9%) in patients with common sepsis.
- **Long ICU stay** (6.2 +- 5.2 days vs 4.4 +- 3.7 days)
- **Difficulty in prediction**
: Symptoms may be more indirect than ARDS initiated in the lungs, so early risk detection through initial time series data has a decisive effect on patients' survival rates.

ARDS Diagnostic Criteria : Berlin Criteria and Application

Based on Berlin Criteria, the international standard diagnostic criterion for ARDS, it has been numerically redefined as follows, taking into account the limitations of the processed MIMIC-IV dataset.

1. 'Oxygenation Indicator'

- based on the most important indicator, PF Ratio Reference numerals
: $\text{PaO}_2 / \text{FiO}_2 \leq 300\text{mmHg}$
- data processing peculiarities
: If FiO_2 missing values are replaced by 0 during preprocessing, an error occurs in which PF Ratio is calculated to infinity.
To prevent this, we modified the logic to determine only $0 < \text{PF Ratio} \leq 300$ as ARDS, increasing the reliability of the data.

2. 'Mechanical Ventilation'

- represents one of the prerequisites for the Berlin Criterion, the $\text{PEEP (Positive Air Pressure)} \geq 5\text{cmH}_2\text{O}$.
- Application Method
: Check if $\text{MechVent} = 1$
- Reason
: Being mechanically ventilated in ICU data means that there is a certain level of respiratory assistance (including PEEP) clinically, so we used this as a basis for our judgment.

Ch 03. Annotation & Labeling

OUR CORE: Labeling process with 'clinical effectiveness'

- **Labeling:** Calculated the oxygenation index (PF - Ratio) taking into account the specificity of medical data.

1. $\text{PaO}_2/\text{FiO}_2 \leq 300$ mmHg
2. $\text{mechvent} = 1$



Process: $\text{mean_PaO}_2 + \text{mean_FiO}_2$
→ generate 'PaO₂/FiO₂' column

Justification (PF - Ratio to 0)

- An error occurs in which the denominator becomes 0 during calculation and $-\infty$ or ∞ occurs because FiO2 is replaced by 0 during preprocessing.
- To address this, force the case to 0. Critical logic: $0 < \text{PF - Ratio} \leq 300$ condition is applied in actual ARDS diagnosis to fundamentally block the misclassification of patients (without data) replaced by 0 above as ARDS patients.



```
# ARDS(급성 호흡곤란 증후군) 라벨링
# 진단 기준:
# -  $0 < \text{PaO}_2/\text{FiO}_2 \leq 300$ 
# - MechVent = 1

def ards_annotation(df: pd.DataFrame):
    # PaO2/FiO2 계산
    ratio = df['mean_PaO2'] / df['mean_FiO2']

    # NaN, inf, -inf → 0 처리
    ratio = ratio.replace([np.inf, -np.inf], 0).fillna(0)

    df['pf_ratio'] = ratio

    # ARDS 진단 기준 적용
    # 진단 기준:
    # -  $0 < \text{PaO}_2/\text{FiO}_2 \leq 300$ 
    # - MechVent = 1
    df['ards_now'] = np.where(
        (df['pf_ratio'] > 0) & (df['pf_ratio'] ≤ 300) & (df['MechVent'] == 1),
        1,
        0
    )

    return df
```

Generating a label that predicts the occurrence of a 'future' at this point in time

- Labeling Strategy

`ards_next_12h`: 1 if the ARDS diagnostic criteria $0 < PF \leq 300$ & `MechVent = 1` are met at least once `within 12 hours` of the current timestamp, or 0.

```

● ● ●

# 12시간 이내 발생 라벨 생성

def ards_prediction_labeling(df: pd.DataFrame,
                             resampling_time: int = 3600,
                             prediction_hour: int = 12,
                             label_col_name: str = 'ards_next_12h',
                             annot_col_name: str = 'ards_now'):

    def _labeling(x: pd.DataFrame):
        # 같은 stay_id 내에서 timestamp 순서대로 정렬
        x = x.sort_values('timestamp').reset_index(drop=True)

        # 현재 row 기준으로, 다음 prediction_window개의 row 안에
        # annot_col_name(SIRS)이 1이 한 번이라도 나오면 label=1, 아니면 0
        x[label_col_name] = [
            1 if x[annot_col_name].iloc[idx+1 : idx+1+prediction_window].sum() >= 1 else 0
            for idx, _ in enumerate(x[annot_col_name])
        ]

        # 마지막 prediction_window개의 row는 미래 정보가 없으므로 버림
        return x.iloc[:-prediction_window, :]

    # '몇 개의 타임스탬프를 미리 볼 것인지'를 index 단위로 변환
    # 예: resampling_time=6시간(21600초), prediction_hour=12 → prediction_window=2 (다음 2 스텝)
    global prediction_window
    prediction_window = int(60 * 60 * prediction_hour / resampling_time)

    labeled_df = df.groupby('stay_id', group_keys=False).apply(_labeling)

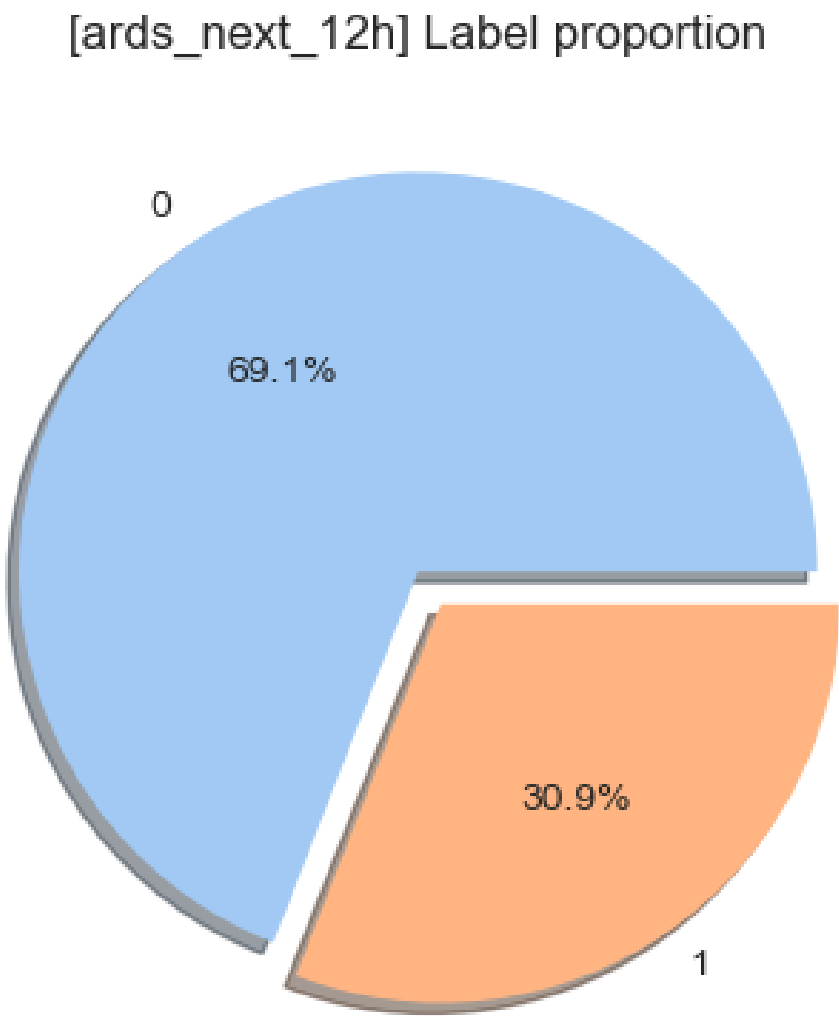
    return labeled_df

```

Justification

- Future Window Trimming
the last part of each patient's time series cannot verify future 12-hour data.
→ delete the last two unpredictable timestamps.

stay_id timestamp				stay_id timestamp		
0	132539	0	→	0	132539	0
1	132539	1		1	132539	1
2	132539	2		2	132539	2
3	132539	3		3	132539	3
4	132539	4		4	132539	4
5	132539	5		5	132539	5
6	132539	6		0	132540	0
7	132539	7		1	132540	1



→ Labeling Result

Justification

- Delete any newly created columns in the labelling process that will not be used to learn the model



Labeling에 사용한 Annotation은 삭제 후 저장

```
annot_list = ['pf_ratio', 'ards_now']
```

```
preprocessed_df.drop(columns = annot_list).to_csv('preprocessed_df_with_label.csv', index = False)
```

Ch 04. Model design and training

Machine Learning Models

Evaluation of Machine Learning Models

⇒ XGB, LGB, RF, DT, LR, NB

Sequence

- ① Use the file 'split_stay_id.json' to split data into train, valid, test sets
- ② Normalization is required to train data on linear family models such as 'lr' and 'nb'.
Prior to normalization, exclude discrete, binary, and static variables
- ③ Normalization using Min-Max scaling
- ④ Exclude columns not used for learning

Deep Learning-Modeling

Evaluation of deep-learning models

⇒ LSTM, TRANSFORMER

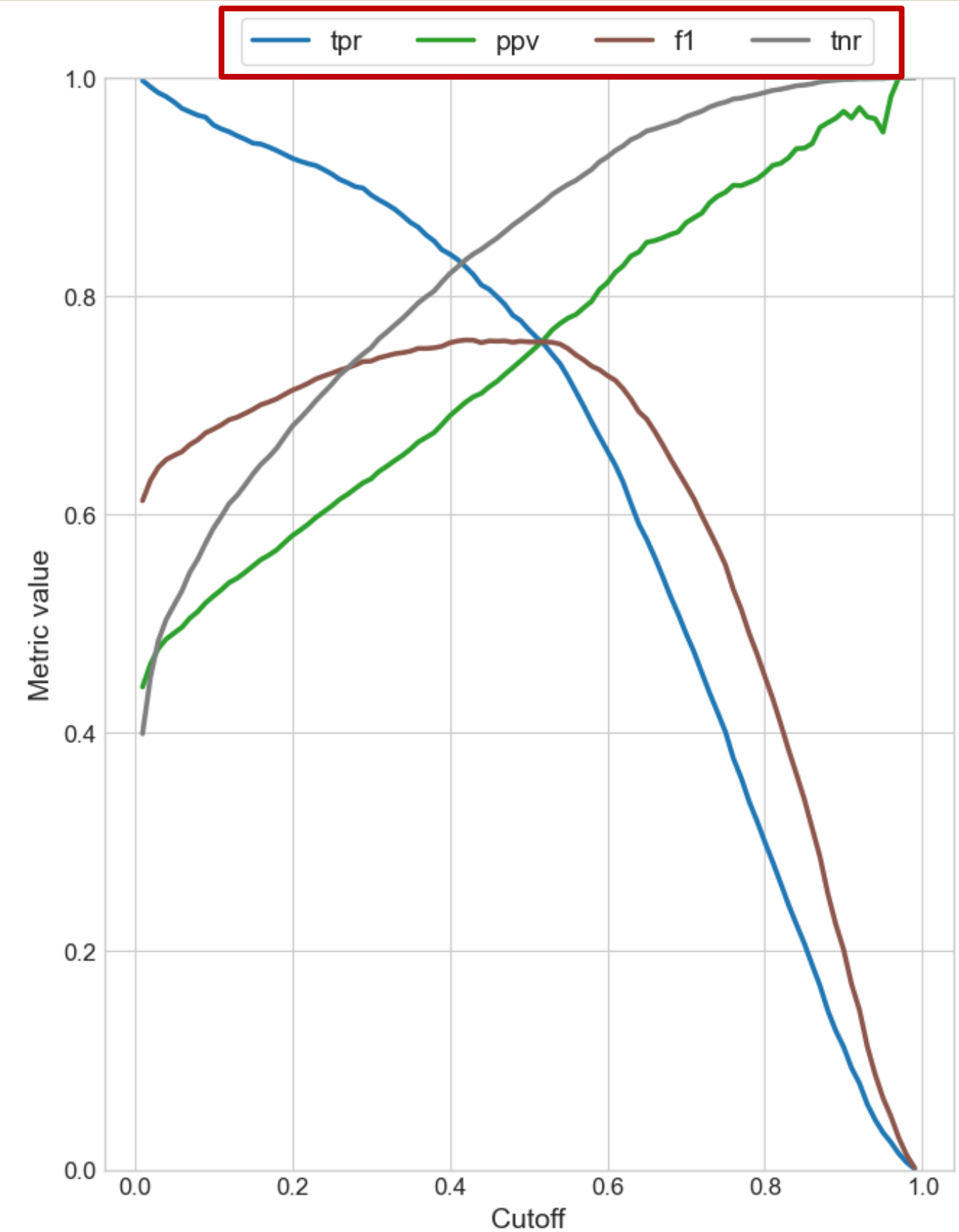
Sequence

- ① Use the file 'split_stay_id.json' to split data into train, valid, test sets
- ② As with ML models, exclude discrete, binary, and static variables, and normalize using Min-Max scaling
- ③ Exclude columns not used for learning + convert to Tensor

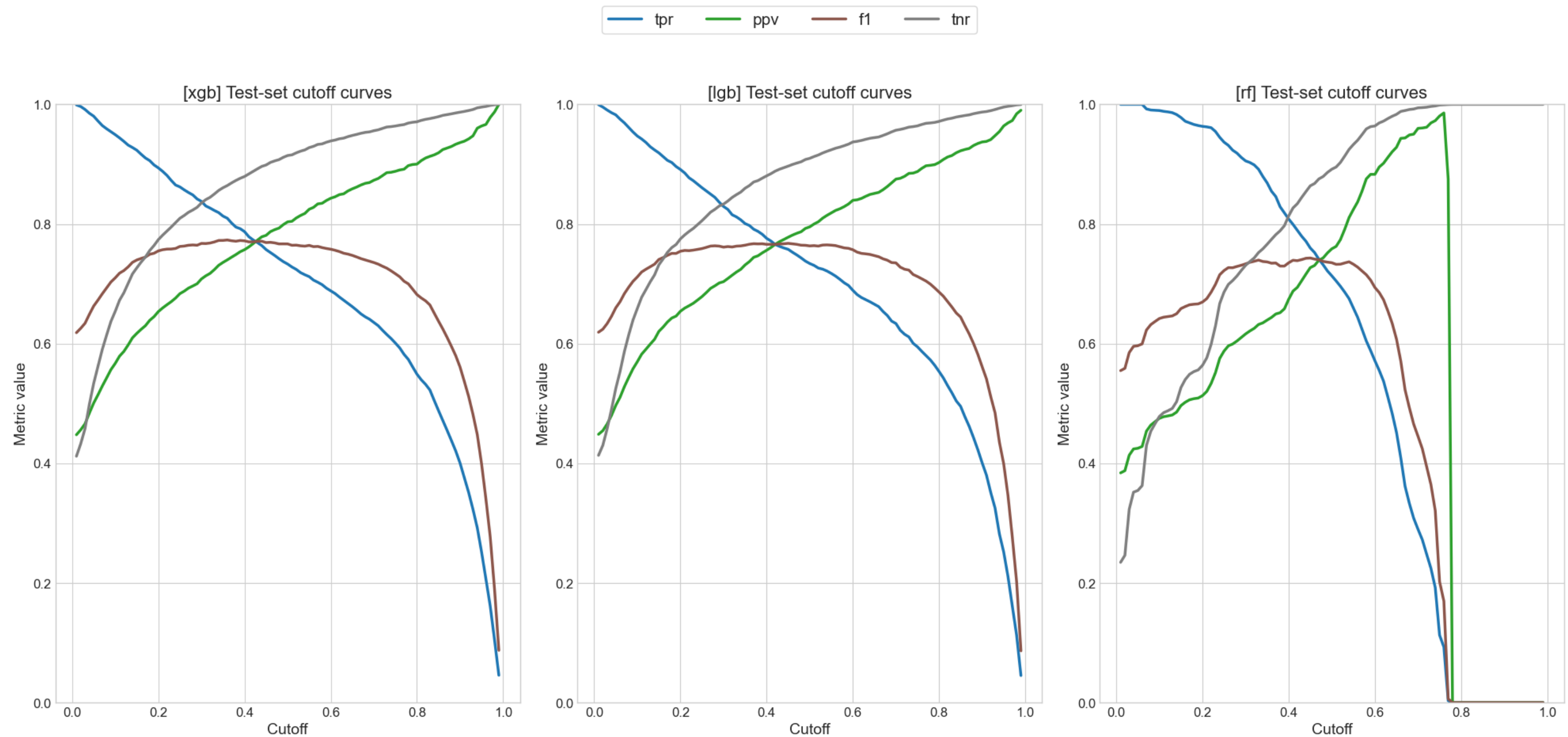
Ch 05. Performance comparison and conclusion

Machine Learning Result

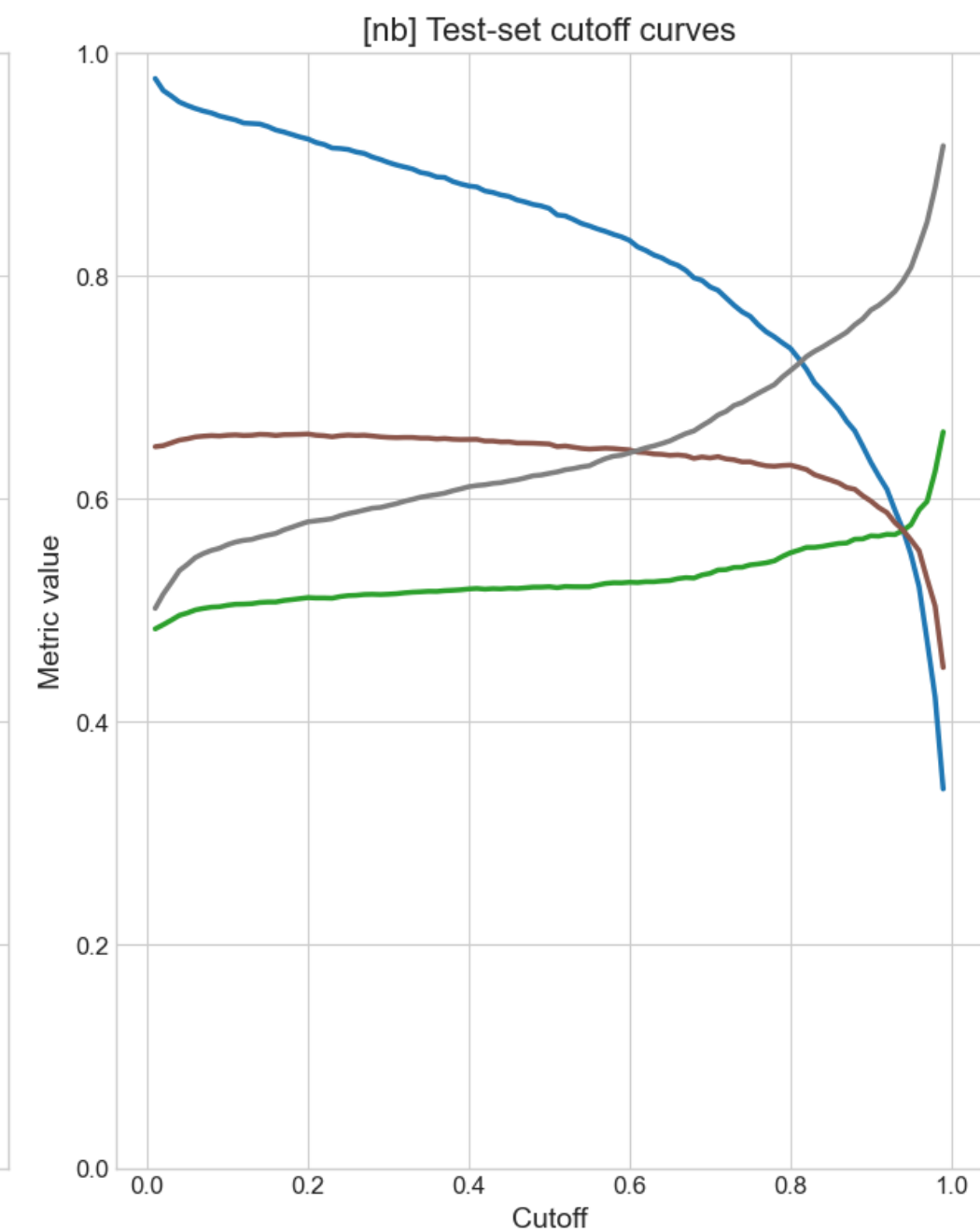
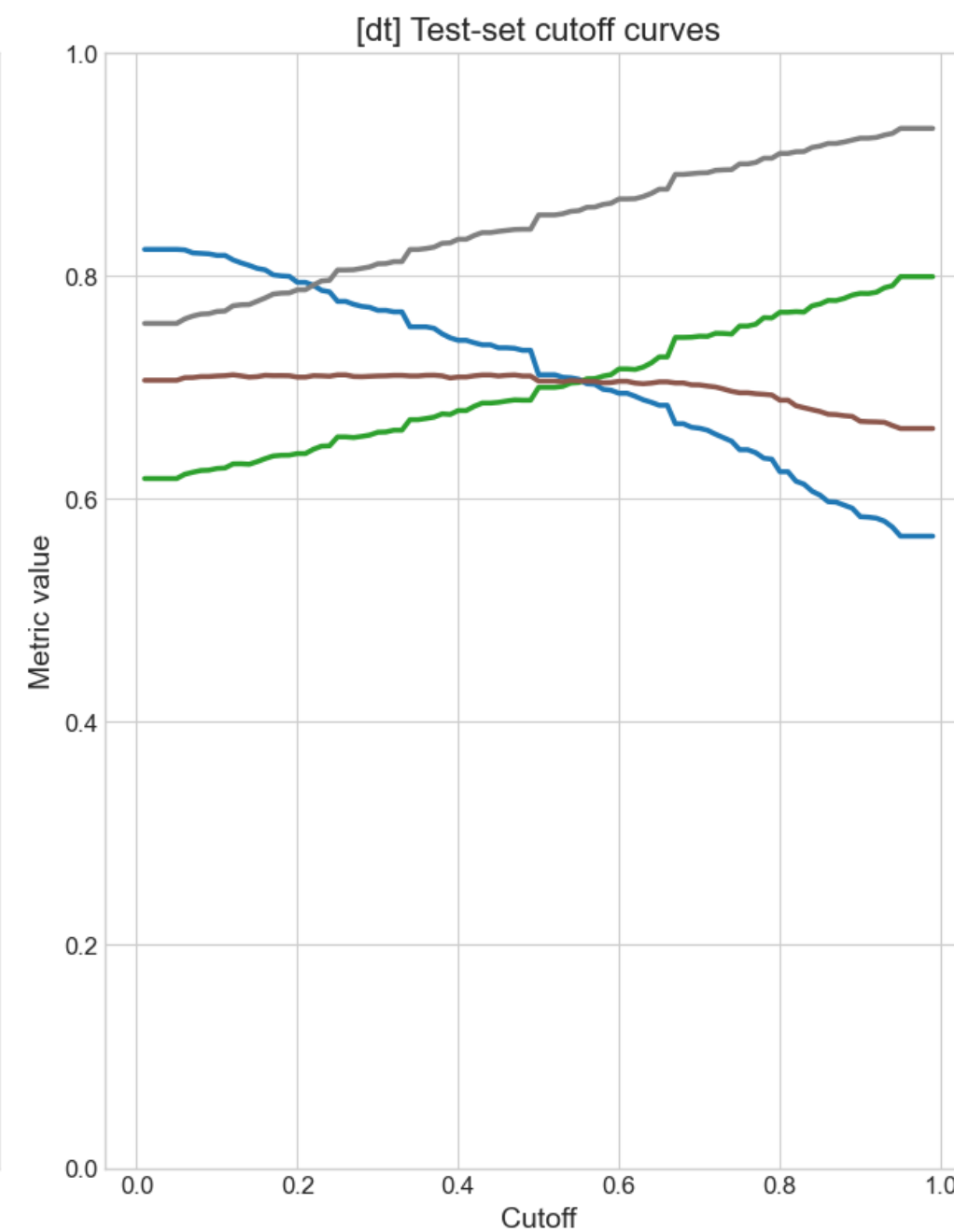
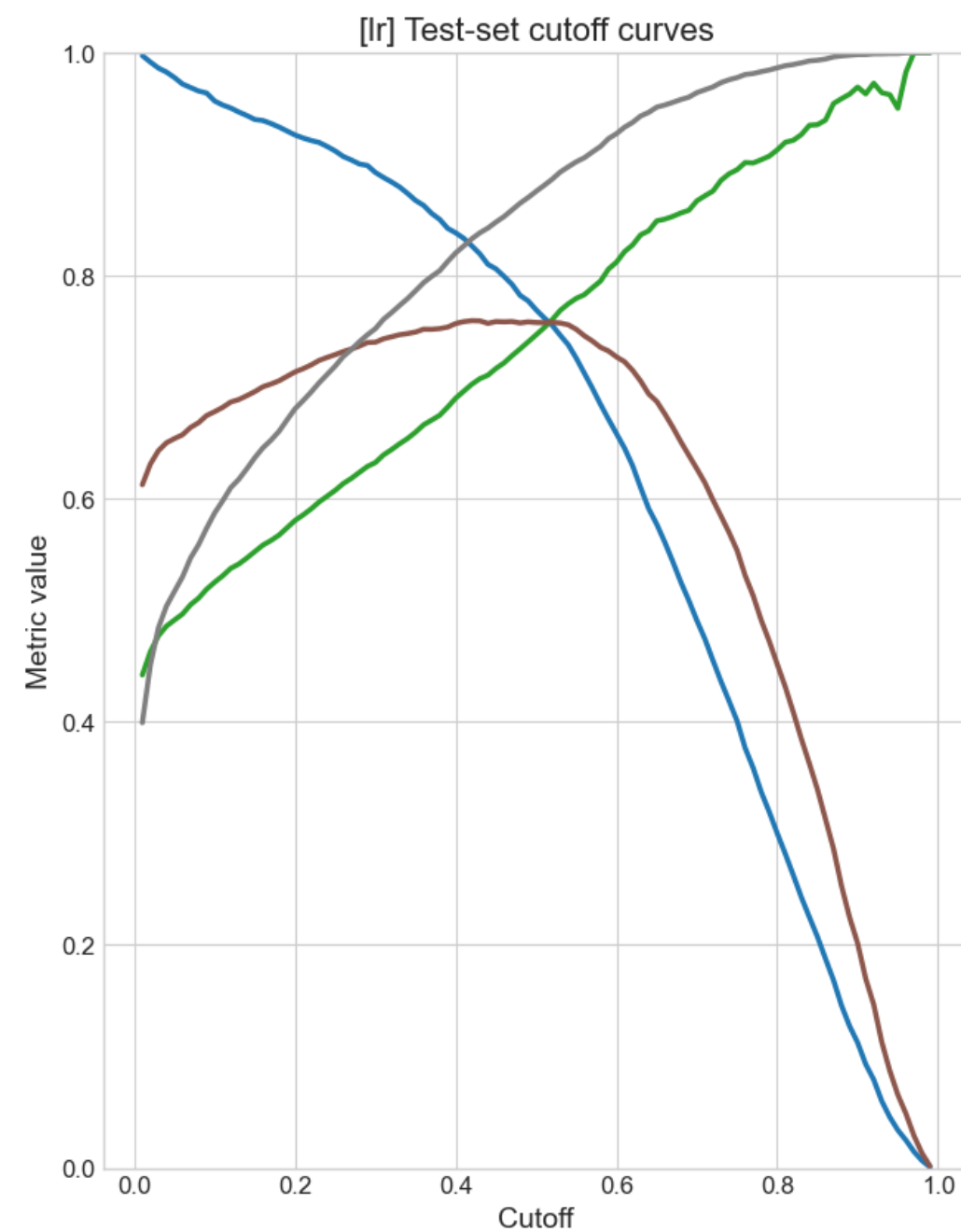
- tpr (= recall) ~ lower type II error
- ppv(= precision) ~ trade-off relation with recall
- F1 score ~ balance of precision and recall
- tnr(= specificity) ~ lower type I error



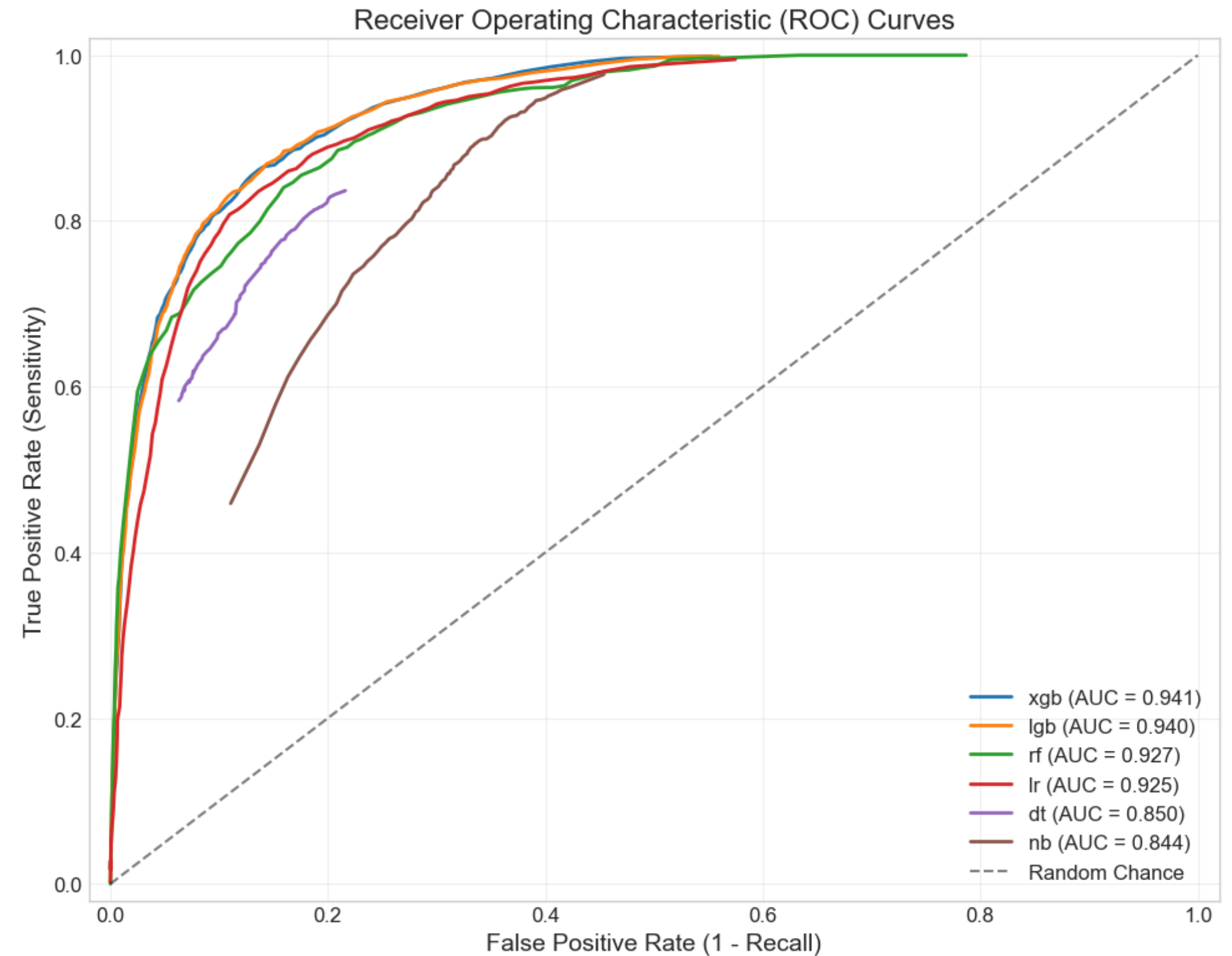
Machine Learning Result



Machine Learning Result



Machine Learning Result



Performance Result Table

Analyzing key performance metrics

- **F1-Score (Optimal Model)**
: LSTM model (0.867) has the highest F1-Score, with the most balanced performance between precision (PPV) and recall (TPR).
- **ROC-AUC**
: Transformer model was the highest at 0.949, while LSTM showed very similar levels at 0.941.
This means that both models have very good predictive power.

Analysis of performance

	tpr	ppv	f1	tnr
model				
dt	0.842154	0.627248	0.718985	0.773148
lgb	0.999536	0.438493	0.609570	0.419823
lr	0.994893	0.432929	0.603322	0.409301
nb	0.976787	0.490557	0.653112	0.540194
rf	1.000000	0.369278	0.539376	0.225800
xgb	0.999071	0.436511	0.607566	0.415404

```
Epoch 30/50 - Training: 100%|██████████| 576/576 [00:01<00:00, 308.4  
=== Epoch 30/50 | Train Loss: 0.2330 | Valid Loss: 0.2749 ===  
Valid Acc: 0.8914 | Precision: 0.8695 | Recall: 0.8648 | F1: 0.8671  
AUROC: 0.9417 | AUPRC: 0.8787  
=====
```

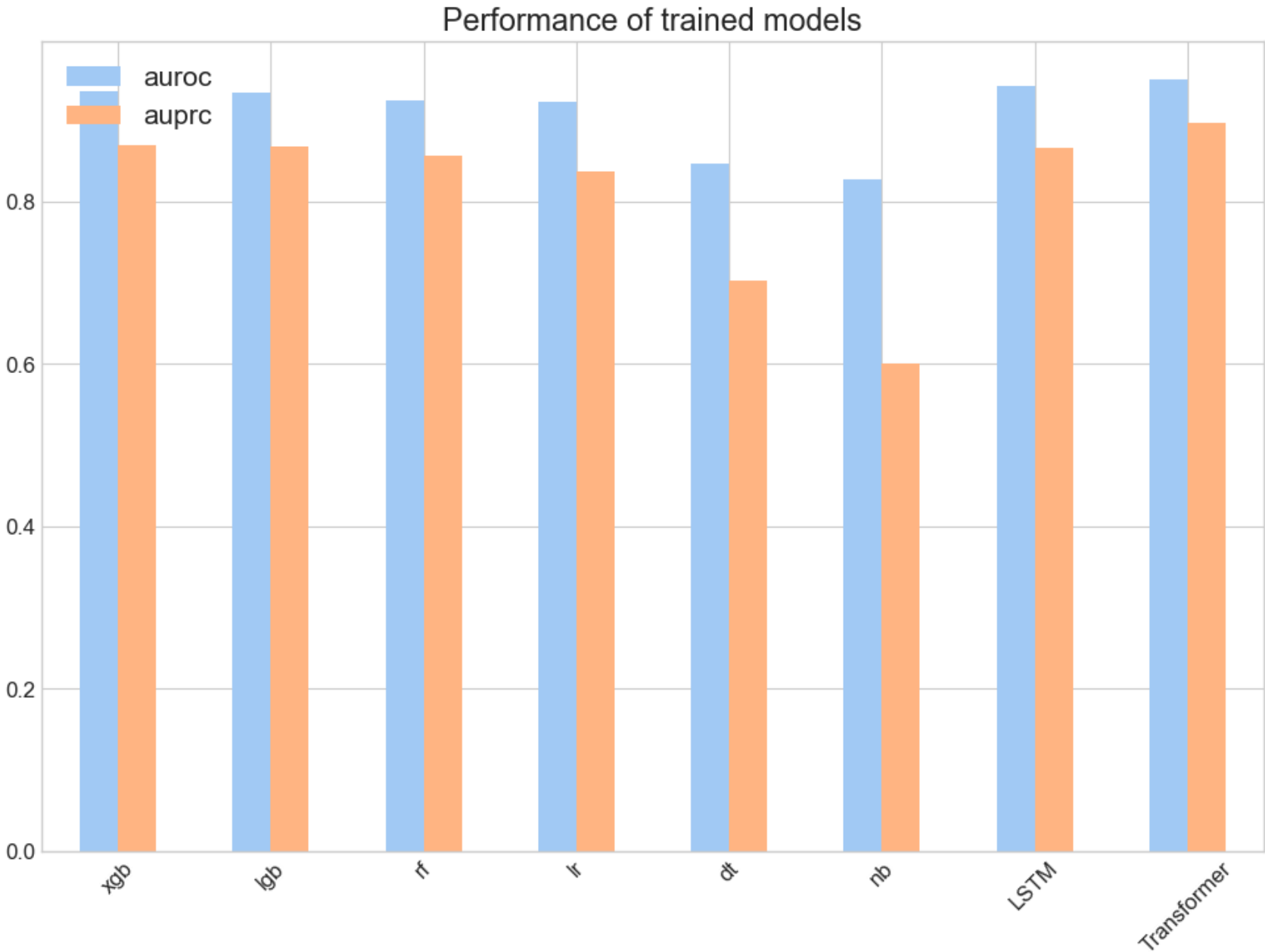
```
----- Test Set Results -----  
Inference: 100%|██████████| 72/72 [00:00<00:00, 406.60  
Test Loss: 0.2609 | Accuracy: 0.8880  
Precision: 0.8687 | Recall: 0.8559 | F1 Score: 0.8619  
AUROC: 0.9483 | AUPRC: 0.8940
```

Graph of performance

	model	auroc	auprc		model	auroc	auprc
0	xgb	0.935761	0.869810	0	xgb	0.940506	0.871924
1	lgb	0.934171	0.866958	1	lgb	0.940136	0.873239
2	rf	0.923695	0.856345	2	rf	0.926561	0.859383
3	lr	0.921995	0.836706	3	lr	0.924685	0.837389
4	dt	0.846115	0.702447	4	dt	0.850477	0.705714
5	nb	0.826414	0.599784	5	nb	0.844159	0.643766
6	LSTM	0.941380	0.865641	6	LSTM	0.941044	0.862550
7	Transformer	0.949389	0.896651	7	Transformer	0.948326	0.893965

CPU

GPU + CUDA 12.8



Conclusion

Core Result

Deep learning models most effectively learns the temporal patterns of time series biosignal data, achieving higher F1-Score and Sensitivity (TPR) in predicting ARDS occurrence than other machine learning models. (including XGBoost and LightGBM)

Clinical Significance

Early intervention and mortality reduction

Predictive models detect risk 12 hours before ARDS occurrence, enabling preemptive clinical interventions such as ventilator application and fluid therapy regulation.

Increase of resource efficiency

By accurately screening ARDS high-risk patients, medical resources (human resources, equipment) can be concentrated and used efficiently.

Missing Value Imputation

1. Originally, the missing values were removed from preprocessing before scaling. This causes distortion in the scaling process, a severe issues in data like Cardiac Arrest.
2. Height >> Gender-Specific Median Imputation, Since height follows a bell curve, but human data has outliers. Median is more robust than mean in terms of outliers

References

https://www.bing.com/images/search?view=detailV2&ccid=37nr9qv1&id=E93505C48FCEFF504EE863C0F15C08E0EF9126912&thid=OIP.37nr9qv1FQ5Bqwg5YRUv5gHaEK&mediaurl=https%3a%2f%2fwww.svhlunghealth.com.au%2fImages%2fUserUploadedImages%2f3460%2f11_SVH_Lung_Health_ARDS_final_1080p.jpg&exph=1080&expw=1920&q=acute+respiratory+distress+syndrome&FORM=IRPRST&ck=209732EB6D6117940F759D2175028855&selectedIndex=0&itb=0&adlt=strict&adlt=strict&idpp=overlayview&ajaxhist=0&ajaxserp=0

https://link.springer.com/article/10.1186/s13054-021-03566-w
https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/74d782e6-5b2e-4443-af95-f974dbff1862/13054_2021_Article_3566.pdf?table=block&id=2bf95538-916b-803d-a44a-c1ebe31c066c&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=aPZTW6qzMdTcypcyIORuNB5TL-OUIKg5qCdQuOZOCzw&downloadName=ARDS+%EC%A7%84%EB%8B%A8+%EA%B8%B0%EC%A4%80+%EB%85%BC%EB%AC%B8.pdf

https://www.nature.com/srep/
https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/904611d9-338d-4d32-9a8d-cbf690c08955/s41598-024-79899-7.pdf?table=block&id=2bf95538-916b-80e1-be8a-cd3a7adf4426&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=WH2DCYP6kfIDseG7zak5qxi9684B39Z-GHAYV491slU&downloadName=%ED%8F%90%ED%98%88%EC%A6%9D+%EB%8F%99%EB%B0%98%ED%95%9C+%ED%8F%90%EC%99%B8%EC%84%B1+ARDS+%EB%85%BC%EB%AC%B8.pdf

https://dl.acm.org/doi/pdf/10.1145/3368555.3384469

https://github.com/MLforHealth/MIMIC_Extract/blob/master/resources/variable_ranges.csv

https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/bb4393e2-c969-4b6f-8835-8695ac79fa3c/MIMIC-IV_%EB%A1%9C_%ED%95%99%EC%8A%B5%ED%95%9C_%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4_%EC%82%AC%EB%A7%9D_%EC%98%88%EC%B8%A1_%EB%AA%A8%EB%8D%B8%EC%9D%98_%EC%84%B1%EB%8A%A5_%EC%99%B8%EB%B6%80_%EA%B2%80%EC%A6%9D.pdf?table=block&id=2b895538-916b-80a0-8550-c93b9bd4f6bb&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=4UsK9CITkaPpVm3D7mJCSIHUQBp75Cc26pLTui2Hoo8&downloadName=MIMIC-IV+%EB%A1%9C+%ED%95%99%EC%8A%B5%ED%95%9C+%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4+%EC%82%AC%EB%A7%9D+%EC%98%8%EC%B8%A1+%EB%AA%A8%EB%8D%B8%EC%9D%98+%EC%84%B1%EB%8A%A5+%EC%99%B8%EB%B6%80+%EA%B2%80%EC%A6%9D.pdf

https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/b02cccfb-2a79-4a8f-b315-bfc11c1df47b/MIMIC-IV_%EB%8D%B0%EC%9D%B4%ED%84%B0%EB%A5%BC.%ED%99%9C%EC%9A%A9%ED%95%9C.%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D_%EA%B8%B0%EB%B0%98.%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4_%EC%9E%AC%EC%9E%85%EC%8B%A4_%EC%98%88%EC%B8%A1_%EB%B0%8F_%EC%A7%91%EC%A4%91_%EC%B9%98%EB%A3%8C_%ED%99%98%EC%9E%90_%EB%B6%84%EB%A5%98.pdf?table=block&id=2b895538-916b-80da-831a-cb3e9b9f5a46&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=2HWVial2wAmJu5DqFGVptcE35HXmdl4Ba-KKRMQQ8Uw&downloadName=MIMIC-IV+%EB%8D%B0%EC%9D%B4%ED%84%B0%EB%A5%BC+%ED%99%9C%EC%9A%A9%ED%95%9C+%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D+%EA%B8%B0%EB%B0%98+%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4+%EC%9E%AC%EC%9E%85%EC%8B%A4+%EC%98%88%EC%B8%A1+%EB%B0%8F+%EC%A7%91%EC%A4%91+%EC%B9%98%EB%A3%8C+%ED%99%98%EC%9E%90+%EB%B6%84%EB%A5%98.pdf

https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/af347ffc-c46a-4012-b9fe-6949543ba1d2/DeepTRACK_%EB%94%A5%EB%9F%AC%EB%8B%9D_%EA%B8%B0%EB%B0%98.%EC%A4%91%ED%99%98%EC%9E%90_%ED%87%B4%EC%8B%A4%EC%97%90_%EB%94%B0%EB%A5%B8.%EC%83%81%ED%83%9C_%EC%95%85%ED%99%94.%EC%9C%84%ED%97%98_%EC%98%88%EC%B8%A1_%EC%8B%9C%EC%8A%A4%ED%85%9C.pdf?table=block&id=2b895538-916b-808d-9caf-f8d99b9b41d0&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=-HLCRQELObp44HjHpTbOKOoB0oGrEAMEUxepbTalc54&downloadName=DeepTRACK+%EB%94%A5%EB%9F%AC%EB%8B%9D+%EA%B8%B0%EB%B0%98+%EC%A4%91%ED%99%98%EC%9E%90+%ED%87%B4%EC%8B%A4%EC%97%90+%EB%94%B0%EB%A5%B8+%EC%83%81%ED%83%9C+%EC%95%85%ED%99%94+%EC%9C%84%ED%97%98+%EC%98%88%EC%B8%A1+%EC%8B%9C%EC%8A%A4%ED%85%9C.pdf

https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/ad301c88-23f2-40e3-8cbc-1c1c638b26dd/%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4_%ED%8C%A8%ED%98%88%EC%A6%9D_%EC%82%AC%EB%A7%9D_%EC%98%88%EC%B8%A1%EC%9D%84_%EC%9C%84%ED%95%9C_%EB%8B%A4%EC%A4%91_%EC%A1%B0%ED%95%A9_%EC%8B%9C%EA%B0%84%EA%B2%A9_%ED%8A%B8%EB%9E%9C%EC%8A%A4%ED%8F%AC%EB%A8%B8_%EB%AA%A8%EB%8D%B8.pdf?table=block&id=2b895538-916b-8036-b2a5-f90fa2c48c65&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=tjo73JLGpGd5Kx7rfzmDu-7N1BjpfY7mNTA9It0JQqs&downloadName=%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4+%ED%8C%A8%ED%98%88%EC%A6%9D+%EC%82%AC%EB%A7%9D+%EC%98%88%EC%B8%A1%EC%9D%84+%EC%9C%84%ED%95%9C+%EB%8B%A4%EC%A4%91+%EC%A1%B0%ED%95%A9+%EC%8B%9C%EA%B0%84%EA%B2%A9+%ED%8A%B8%EB%9E%9C%EC%8A%A4%ED%8F%AC%EB%A8%B8+%EB%AA%A8%EB%8D%B8.pdfhttps://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/8c52c381-5fef-4f50-a5f8-dcb725a96776/%EC%A0%84%EC%9E%90%EA%B1%B4%EA%B0%95%EA%B8%B0%EB%A1%9D_%EB%8D%B0%EC%9D%B4%ED%84%B0%EC%97%90%EC%84%9C%EC%9D%98_%ED%8F%90%EB%A0%B4_%EC%95%84%EC%A2%85%EC%97%90_%EB%8C%80%ED%95%9C_%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D_%EA%B8%B0%EB%B0%98_%EC%B0%A8%EC%9D%B4_%EB%B6%84%EC%84%9D.pdf?table=block&id=2b895538-916b-800f-ba0e-cc948cdae744&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=BIIP8G0-FLgyJ3hTV7Gvdyw3B7ER6gEXq-x_bNmy7as&downloadName=%EC%A0%84%EC%9E%90%EA%B1%B4%EA%B0%95%EA%B8%B0%EB%A1%9D+%EB%8D%B0%EC%9D%B4%ED%84%B0%EC%97%90%EC%84%9C%EC%9D%98+%ED%8F%90%EB%A0%B4+%EC%95%84%EC%A2%85%EC%97%90+%EB%8C%80%ED%95%9C+%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D+%EA%B8%B0%EB%B0%98+%EC%B0%A8%EC%9D%B4+%EB%B6%84%EC%84%9D.pdf

https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/8d01e600-4fda-453f-a5bd-d67aa9746ee0/%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D%EC%9D%84_%ED%99%9C%EC%9A%A9%ED%95%9C_%EB%8B%B9%EB%87%A8%EB%B3%91_%ED%99%98%EC%9E%90%EC%9D%98_%EA%B4%80%EC%83%81_%EB%8F%99%EB%A7%A5_%EC%A7%88%ED%99%98_%EB%AA%A8%EB%8D%B8_%EA%B0%9C%EB%B0%9C.pdf?table=block&id=2b895538-916b-80bc-9582-c63e44551a9d&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=wDJRjBggmQFUKn8cfWQt3jt9irgCpL3s29rpviHLzs&downloadName=%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D%EC%9D%84+%ED%99%9C%EC%9A%A9%ED%95%9C+%EB%8B%B9%EB%87%A8%EB%B3%91+%ED%99%98%EC%9E%90%EC%9D%98+%EA%B4%80%EC%83%81+%EB%8F%99%EB%A7%A5+%EC%A7%88%ED%99%98+%EB%AA%A8%EB%8D%B8+%EA%B0%9C%EB%B0%9C.pdf

https://file.notion.so/f/f/13695538-916b-811a-ad55-0003e42c7a54/0fefa5f0-3500-4af0-bcf5-e3de947521c4/%EC%A0%84%EC%9E%90%EC%9D%98%EB%AC%B4%EA%B8%B0%EB%A1%9D_%EB%B9%85%EB%8D%B0%EC%9D%B4%ED%84%B0%EC%97%90%EC%84%9C%EC%9D%98_%ED%86%B5%EA%B3%84_%EB%B0%8F_%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D_%EB%B0%A9%EB%B2%95%EC%9D%84_%ED%99%9C%EC%9A%A9%ED%95%9C_%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4_%EC%82%AC%EB%A7%9D_%EC%9C%84%ED%97%98_%EC%8A%A4%EC%BD%94%EC%96%B4%EB%A7%81_%EC%8B%9C.pdf?table=block&id=2b895538-916b-809f-b7f1-ff76767991a0&spaceld=13695538-916b-811a-ad55-0003e42c7a54&expirationTimestamp=1765670400000&signature=G5yQOWRP-ZrWPJs-raT6eSMuZU7GCm_SUGFKliaaPH0&downloadName=%EC%A0%84%EC%9E%90%EC%9D%98%EB%AC%B4%EA%B8%B0%EB%A1%9D+%EB%B9%85%EB%8D%B0%EC%9D%B4%ED%84%B0%EC%97%90%EC%84%9C%EC%9D%98+%ED%86%B5%EA%B3%84+%EB%B0%8F+%EB%A8%B8%EC%8B%A0%EB%9F%AC%EB%8B%9D+%EB%B0%A9%EB%B2%95%EC%9D%84+%ED%99%9C%EC%9A%A9%ED%95%9C+%EC%A4%91%ED%99%98%EC%9E%90%EC%8B%A4+%EC%82%AC%EB%A7%9D+%EC%9C%84%ED%97%98+%EC%8A%A4%EC%BD%94%EC%96%B4%EB%A7%81+%EC%8B%9C.pdf

THANK YOU